

PROJECT HELM

Helm Phase 0: Solution Architecture & MVP Delivery Plan

Client meeting, 9 October 2026

Cyclone

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Contents

How to read this document	2
Part I: Business and client	3
1 Executive overview	3
2 Confirmed stack: what each system provides	5
3 The Member journey	6
4 Phase 0 scope	7
5 Feasibility summary	8
6 MVP delivery plan	9
7 Security and compliance summary	11
8 Decisions needed at the meeting	12
Part II: Technical implementation	13
9 Traceability to Terrel's requirements	13
10 Refined high-level PRD	15
11 Solution architecture	18
12 End-to-end data flows	21
13 Genealogy, Commission and Phase 1 readiness	26
14 Backlog, dependencies and schedule	28
15 Launch and operational readiness	29
16 Implementation decisions and feasibility gaps	31
Appendix A: Evidence	32
Appendix B: Assumptions and gaps	33
Appendix C: Questions for providers	34

How to read this document

Part I (sections 1–8) is for Business and the client. It explains what Members can do, which platform supplies each capability, what Cyclone builds, how fast an MVP can credibly launch and which decisions are needed at the meeting. It does not require the technical sections.

Part II (sections 9–16) is for the Tech teams. It refines the PRD and specifies the architecture, data flows, genealogy model, backlog, launch gate and responsibilities.

The appendix holds evidence links, assumptions, unresolved details and questions for providers.

Labels used throughout:

Label	Meaning
Verified	Read in the provider's own documentation, API specification, contract source or pricing page on 8 October 2026 (links in Appendix A).
Correspondence	Stated in provider correspondence supplied by Business; not independently verified.
Recommended	Cyclone's proposed design or default. Not yet approved.
TBD	Assumption or open item. Needs a decision or provider confirmation.

The stack is confirmed: **PillarsHub** (MLM core), **Privy** (login and wallets), **Enzyme Onyx** (vault infrastructure), **Chatwoot** (support), Cyclone's **Tap Prediction game and points**, and a custom **Helm Web App** with its backend. This document does not compare or reselect providers. Terrel's requirements (HELM Community Technology Requirements v0.2, 6 October 2026) are traced in section 9.

PART I: BUSINESS AND CLIENT

1 Executive overview

Goal of Phase 0: capital is waiting to be deployed, so the MVP is cut to the fastest path that lets Business **onboard Members, let them join the vault (referred to by Business as “staking”), capture the genealogy correctly, and pay a simple referral Commission.** Everything else waits.

The confirmed stack can deliver this. Each capability is available from a confirmed provider or needs only configuration and a contained amount of Cyclone integration work. No evidenced blocker requires changing a provider. Terrel’s requirements (v0.2, 6 October 2026) have been mapped to this scope in section 9.

What Members will be able to do at launch:

1. Join Helm through an invite link, or be imported from the alpha community, with their referral history kept.
2. Log in with email or social login and receive a Helm wallet automatically. Only the Member can move its funds.
3. Fund that wallet by sending USDC on one network from any exchange or wallet.
4. Read the vault terms, deposit into the approved vault, see their confirmed position, and request a Redemption.
5. See their direct team and referral link, and, as Builders, see Commissions as pending, held or paid.
6. Receive Commission payouts in USDC directly to their Helm wallet, in batches approved by Finance.
7. Play the existing Tap Prediction game and earn Points, which have no cash value.
8. Chat with support from inside the app; support sees who they are and their status, never their keys.

Why this is the fastest credible path:

- **Cyclone builds only the glue:** the member experience, one backend that connects the providers, a ledger of identities, referrals and events, the payout executor, and the reconciliation and admin views that make every money flow auditable.
- **Each provider does what it already does:** Privy runs login and wallets; PillarsHub stores the genealogy and calculates Commissions with its existing Unilevel plan; Onyx issues vault shares and runs deposit and redemption queues; Chatwoot handles conversations.
- **MLM screens are reused, not rebuilt.** Genealogy browsing, Sponsor and placement edits, plan configuration, bonus review, period close and bonus release are done in the PillarsHub Portal. Builders get the essentials in Helm plus an “Open full back office” link into PillarsHub’s back office through single sign-on. Verified (SSO). Helm builds only what PillarsHub does not have.
- **Payouts reuse PillarsHub’s batch mechanism.** PillarsHub sends approved payout batches to a merchant endpoint that Helm builds; Helm pays USDC from a Business-controlled multisig treasury and reports the result back. Verified. No payment provider is needed.
- **Direct crypto funding** keeps payment-provider onboarding off the critical path.

What can start immediately (no provider or legal answer needed): the Helm Web App shell, Privy login and wallets, the member, referral and event data model, the PillarsHub staging sync, alpha-import tooling (once sample data arrives), the payout executor on testnet, the Chatwoot widget, chain indexing, and the full deposit and redemption journey on Enzyme’s Sepolia test network.

What gates live funds and payouts (none of these blocks the build):

1. **The vault strategy and its Manager.** Onyx is vault infrastructure, not a strategy. Someone must run the strategy, report the vault’s value and return liquidity for Redemptions. Verified.
2. **Vault ownership and Enzyme’s upgrade rights.** The vault Owner and Admins are fully trusted by the protocol, and Enzyme can upgrade vault contracts. The owning entity, a multisig and the upgrade terms in the licence agreement must be settled. Verified.
3. **Enzyme’s licence agreement (MLA) and commercial terms.** Production use requires a commercial agreement. Verified.
4. **The Commission basis for the simple payout.** Which event pays (for example, a confirmed vault deposit, or vault fee revenue), the three level rates, period, hold and minimum must be approved by

Owen, Finance and counsel. Terrel's requirements are explicit that "a raw deposit ... must not silently replace approved eligible activity".

5. **Legal applicability review** of the operating entity, launch countries, KYC tier, vault terms and the referral payout model.
6. **Provider acceptance of the business model**, in writing, from Privy, Enzyme and PillarsHub.

Earliest credible schedule (estimate, not a commitment): an end-to-end test slice including a test payout about 3 weeks after kickoff; a controlled launch with deposits in about **8–12 weeks**, provided vault, licence, Commission-basis and legal answers arrive by around week 6. The first live payout follows the first approved period close and hold. Section 6 gives the assumptions, and section 6.1 sets out what is achievable within 30 days.

2 Confirmed stack: what each system provides

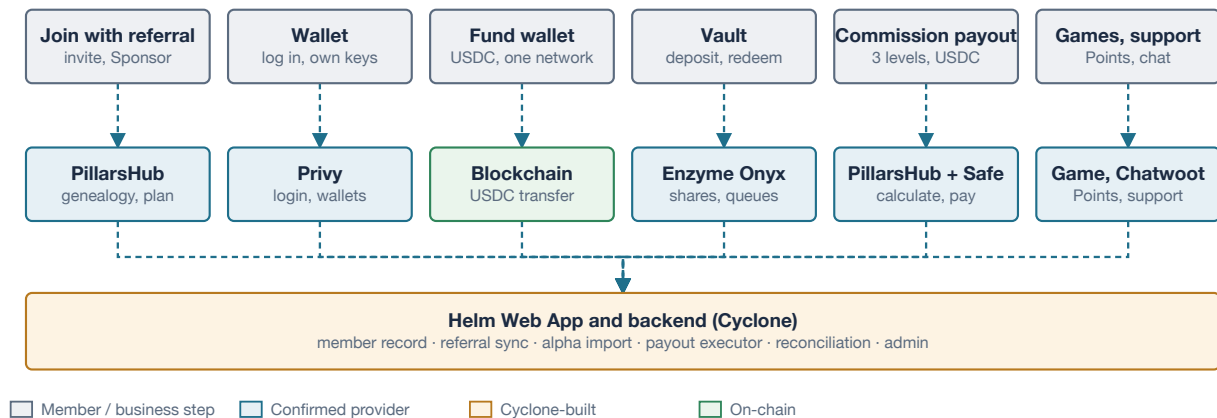


Figure 1: Business capabilities, the platform that provides each, and what Cyclone builds.

Capability	Provided by	Cyclone builds	Status
Join and log in; one stable Helm identity	Privy (login, identity token)	Member record with Customer and Builder status, Privy mapping	Feasible
Wallet	Privy embedded wallet, owned by the Member; gas sponsored by Helm	Wallet screens; balance display	Feasible
Referral, genealogy and alpha import	PillarsHub (customer with enroller, Unilevel tree, full-tree export, dated placements)	Referral capture and validation, sync, alpha import and reconciliation	Feasible with configuration
Wallet funding	Blockchain (USDC transfer to the Member's address)	Address display, confirmation tracking, screening	Feasible
Vault deposit and redemption	Enzyme Onyx (allowlisted deposit queue, redemption queue, shares, public read API)	Terms screens, transaction flows, status tracking, allowlist updates	Feasible with configuration; live use blocked pending strategy, MLA and ownership
Simple Commission payout	PillarsHub (real-time calculation, bonus release, payout batches to a custom merchant)	Eligible-event posting, payout executor, treasury proposals, status callback, statements	Feasible with configuration and custom work; blocked pending approved Commission basis
Team view and Builder back office	PillarsHub back office (tree, reports, bonus detail) through SSO	Essentials in Helm (referral link, team counts, earnings summary, payouts) and the SSO hand-off	Feasible
MLM administration	PillarsHub Portal (genealogy, placements, plan, bonuses, period close, release)	Deep links by Member ID; change detection and reconciliation	Feasible
Games and Points	Cyclone Tap Prediction and points service	Account linking, verified events, Points display	Feasible pending game API documentation
Support	Chatwoot Cloud (web widget with identity validation)	Widget, identity signing, agent context view in Helm admin	Feasible
Operations	Provider dashboards (Onyx Admin App, PillarsHub Portal, Chatwoot)	Helm admin: exceptions, approvals, reconciliation, audit log	Feasible

3 The Member journey

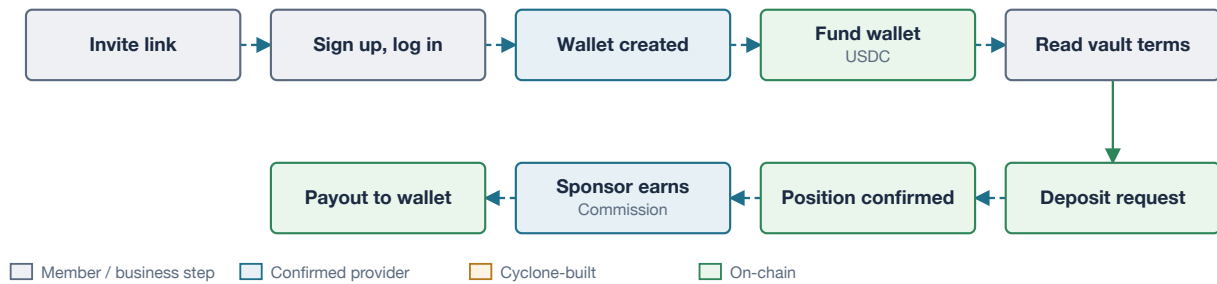


Figure 2: Main Member journey. Wallet funding, vault deposit and Commission are separate events; Points are separate from money.

1. **Invite and signup.** The Member opens a Sponsor's invite link and logs in with Privy. Helm records the Referral Request and confirms it once PillarsHub accepts it. Alpha-community Members are imported with their existing Sponsor.
2. **Wallet.** Privy creates an embedded wallet. Only the Member can sign with it; Helm and Privy cannot move its funds.
3. **Wallet funding.** Helm shows one supported network and asset (recommended: USDC on Arbitrum) and the Member's address. Funding moves money into the Member's own wallet; it is **not** an investment and creates no Commission.
4. **Vault deposit.** The Member reads the vault terms, fees and risks, then signs two transactions: permission for the vault to take the amount, and the deposit request. The request is pending until vault operations process it at the next valuation. Only then does the Member hold vault shares.
5. **Position.** Helm shows the confirmed share balance and its value as of the vault's last valuation, with the valuation date.
6. **Commission.** If the approved basis is met, PillarsHub calculates the Sponsor's Commission (up to three levels). Builders see it as pending, held and then payable.
7. **Payout.** After the period closes and the hold passes, Finance approves the batch. Helm pays USDC from the treasury to each Builder's Helm wallet; the Builder sees it as paid, with the transaction reference.
8. **Redemption and engagement.** Members can request a Redemption, processed when liquidity is available with each stage shown, and play Tap Prediction for Points.

4 Phase 0 scope

In scope (Pilot)	Deferred; stays disabled until approved
One network and one funding asset (recommended: USDC on Arbitrum One)	Integrated purchase or top-up (Paymenture or similar)
One approved Onyx vault, asynchronous deposits, queued Redemptions	Additional chains, assets or vaults; instant deposits
Privy embedded wallets with sponsored gas	Server wallets, automated strategies, policies
Referral capture, Customer and Builder status, full genealogy in PillarsHub, alpha import	Self-service or bulk placement moves; the governed placement window (GEN-04 to GEN-06)
Simple Commission: existing three-level Unilevel plan, one approved eligible event, USDC payout in Finance-approved batches	Ranks, qualification rules, matching or leadership bonuses, campaigns, contests
Referral link, team counts, earnings summary and payouts in Helm; full back office in PillarsHub via SSO	Custom Helm versions of PillarsHub screens; deep team analytics, leader contact tools, CRM
Existing Tap Prediction game and Points (no cash value)	Points redemption for value, new games, tokens
Chatwoot web chat with agent context; FAQ and runbooks	Telegram and WhatsApp support; AI agent
Sanctions screening of wallet addresses; country eligibility; Member terms	Full KYC unless counsel requires it (decision D6)

5 Feasibility summary

Area	Verdict	Key evidence	What remains
Login and identity	Feasible	Privy issues verifiable identity tokens; one Privy user maps to one Helm member. Verified.	Confirm Privy accepts the business model.
Wallet	Feasible	User-owned embedded EVM wallet; Arbitrum supported; native gas sponsorship; key export. Verified.	Losing the only login method loses the wallet; offer a second login method.
Genealogy and alpha import	Feasible with configuration	Customer records take a Helm-defined ID, signup date and enroller at creation; the whole tree can be listed with an as-of date; placements are dated. No bulk import API. Verified.	Alpha records; auto-placement, depth limits and correction method in staging.
Commission calculation	Feasible	Payable depth is set in the plan; custom volume sources carry an external ID; calculation is real time. Verified.	Approved basis, rates, period and hold; plan configured by PillarsHub.
Commission payout	Feasible with custom work	PillarsHub sends release batches to a custom merchant endpoint authenticated by a callback token; the merchant reports “Success”, “Failure” or “Pending” per payment; custom currency codes are allowed. Verified.	Helm builds the merchant endpoint and treasury executor; Finance approval flow.
Wallet funding	Feasible	Standard USDC transfer; Helm indexes the chain.	Choose network and asset.
Vault deposit and redemption	Feasible with configuration; live use blocked	Allowlisted deposit queue; Member must sign; admin executes after valuation; queued Redemptions; read-only API. Verified.	Strategy and Manager, MLA, ownership, upgrade terms, operating process.
Games and Points	Feasible pending evidence	No game documentation supplied.	API documentation, event verification and abuse controls.
Support	Feasible	Web widget with signed identity; Cloud plans from \$19 per agent per month. Verified.	Agent context view; FAQ and runbooks (SUP-04 to SUP-06).
Compliance tooling	Gap within the stack	No confirmed provider offers usable KYC or screening for this model.	Counsel decision; add a screening service at the Helm backend if required.

6 MVP delivery plan

Delivery is organised so that engineering never waits for provider or legal answers. Those answers gate **live funds and payouts**, not the build.

Stage	Duration (estimate)	Outcome	Gate to next stage
0 – Access and decisions	Week 1	Meeting decisions recorded; PillarsHub staging, Privy, Chatwoot, Onyx Sepolia vault, game docs and alpha sample data in hand	Access confirmed
1 – Thin end-to-end slice	Weeks 1–3	On test networks: invite → signup → wallet → referral accepted → funding → deposit → executed position → test Commission → test payout batch → Redemption	Slice demonstrated
2 – MVP build	Weeks 3–8	Member UI with team view and statements, admin approvals, alpha import rehearsal, reconciliation, screening, terms, support context, monitoring	Feature-complete on test networks
3 – Production readiness	Weeks 6–10 (overlaps)	Production vault deployed and handed over; plan configured and proven with Finance examples; treasury multisig set up; security review; legal sign-off; alpha import and payout rehearsal	Launch gate (section 15) passed
4 – Controlled launch	From about week 8–12	Invitation-only cohort with exposure limits; first payout after first period close and hold	Agreed checks pass, then expand

6.1 30-day plan

What can realistically be in place 30 days after kickoff, by how much depends on engineering alone and how much on decisions outside it. All items are estimates for a team of five to six engineers with provider access in week 1.

Tier	What is live at day 30	Depends on
1. Certain (engineering only)	Full journey on test networks: signup → wallet → referral → USDC funding → Vault Deposit → position → Commission calculation → test payout batch from the treasury Safe → Redemption. Alpha import tool run and reconciled on sample data. Admin roles, maker-checker, daily reconciliation, Chatwoot	Provider test access; alpha sample data
2. Likely, if Business decides in week 1	Preregistration live with real users, no money yet: signup, wallet, referral link, genealogy in PillarsHub production, alpha community imported with original Sponsors and dates, direct-team view, support. Matches Terrel's release sequence and ID-04: preregistered people keep attribution and cannot earn	Member terms and privacy notice; allowed countries; alpha records; PillarsHub production slot (Tuesday to Thursday)
3. Possible around day 30–35, only if every external item lands on time	Capped real-funds pilot: an invited cohort (for example alpha leaders) with per-Member and total limits deposits into the production vault	See the conditions below
4. Not realistic within 30 days	First real Commission payout; placement window, ranks and campaigns; full KYC if counsel requires it (adds about 1–2 weeks)	Payout needs an approved basis plus one period close and hold: about day 45–60 with a weekly period and a short hold

Conditions for tier 3:

Condition	Needed by
Vault strategy and Manager appointed; Owner multisig set up	Week 1
Enzyme MLA signed; production vault deployed and handed over	Signed in weeks 1–2; deployed by week 3 (Enzyme publishes no lead time)

Condition	Needed by
Counsel confirms entity, launch countries, KYC tier and vault terms	Week 3
Focused security review of vault configuration, keys and integration	Week 4
Operations rehearsal with a small operator-funded amount	Week 4

Week by week:

Week	Engineering	Business, Finance and counsel
1	Access, environments, Privy login and wallet, member and referral model, PillarsHub staging sync, Onyx Sepolia deposit	Decisions D1–D13; vault strategy and Manager; MLA; alpha records; Commission basis drafting; counsel engaged
2	Thin slice on test networks; alpha import dry run; event pipeline; payout endpoint and Safe proposals on testnet	Member and Builder terms; allowed countries; MLA signature
3	Builder essentials and PillarsHub SSO, admin approvals, screening, support context; preregistration release candidate	Counsel answers; production vault deployment; Commission basis approved
4	Preregistration go-live; alpha cutover; load and recovery checks; security review fixes; real-funds rehearsal	Security review; operations rehearsal; go/no-go for the capped pilot

Recommended day-30 milestone for the meeting: *preregistration live, alpha community imported, genealogy locked, vault and payout proven on test networks.* Run the tier-3 conditions in parallel so real deposits open as soon as they clear, and approve the Commission basis early so the first payout is not delayed further.

6.2 Assumptions and running costs

Assumptions behind these ranges: a Cyclone team of about five to six engineers plus QA and a delivery lead; provider access and alpha sample data in week 1; game API documentation by week 2; vault strategy, MLA, ownership, Commission basis and legal answers by around week 6. PillarsHub only launches clients Tuesday to Thursday, 9:00–16:00 US Mountain Time. Verified. No date in this table is a commitment.

Indicative running costs (USD per month unless stated; prices verified on 8 October 2026 where marked):

Item	Cost	Status
Privy	Free up to 499 monthly active users; \$299 up to 2,499; \$499 up to 9,999. The free tier includes \$1M of transaction volume per month.	Verified; overage and paid-tier volume limits TBD
Gas	Sponsored Member transactions plus treasury payout transactions	TBD (usage-based)
Enzyme Onyx	Public page: \$2,000 per month, or 20% of vault fees with a \$6,000 annual minimum. Correspondence: \$5,000 deployment plus 0.25% of AUM, or 20% revenue share.	Verified (public) vs correspondence; reconfirm in the MLA
PillarsHub	Not published	TBD
Treasury multisig	Safe contracts (gas only)	Recommended
Chatwoot Cloud	\$19 or \$39 per agent (Startups, Business); \$99 for Enterprise with audit logs and SSO	Verified
Sanctions address screening	Free Chainalysis sanctions API at pilot scale	Verified (discovery research)
KYC service, if counsel requires it	About \$0.33–\$1.85 per check at published prices	Verified (discovery research)
Security review of Helm's deployment	Quotation needed	TBD

7 Security and compliance summary

- **Custody.** Member wallets are self-custodial: only the Member signs. Neither Cyclone nor Privy can move Member funds. Verified (Privy). The vault is different: its Owner and Admins can set the share value and withdraw vault assets to the strategy wallet, and Enzyme can upgrade the contracts. Who holds those roles, under what controls, is the main custody question for Business and counsel.
- **Payout treasury.** Commissions are paid from a Business-owned multisig treasury. Helm's backend can only propose payout transactions; Finance signers approve them. Payouts go only to the Builder's own Helm wallet, so no one can redirect a payment by changing a destination (PAY-01).
- **Admission to the vault can be controlled on-chain.** Onyx lets only allowlisted wallet addresses deposit. Helm adds a Member's address only after its eligibility checks pass. Verified.
- **No provider in the stack supplies usable KYC or sanctions screening for this model.** Privy's KYC runs on Bridge, whose terms prohibit MLM. If counsel requires identity checks, a screening service is added at the Helm backend.
- **Minimum before real funds or payouts, whatever counsel concludes:** screening of every Member wallet, funding source and Redemption destination; country eligibility; Member and Builder terms, risk, fee and earnings disclosures, and a privacy notice; a written procedure for flagged users and transactions; payee status required for payouts (PAY-04).
- **Referral payouts on vault activity are the most sensitive legal point.** Paying Commissions on deposits ties rewards to new money coming in; counsel must approve the basis before the first payout. Points have no cash value and are kept separate from money and Commissions.
- **Provider audits do not cover Helm's deployment.** Helm's vault configuration, keys, valuation process, payout executor and integrations need their own security review before funds.

8 Decisions needed at the meeting

#	Decision	Recommended answer	Why it matters
D1	Network and funding asset	Arbitrum One, native USDC; test on Ethereum Sepolia	Fixes wallet, vault, payout and indexing work
D2	Commission basis for the simple payout	One approved eligible event (confirmed vault deposit or vault fee revenue), three Unilevel levels, period, hold and minimum approved by Owen, Finance and counsel; engineering builds the event pipe now	Payout cannot go live without it; the basis drives legal risk
D3	Vault strategy, Manager and ownership	Business appoints the Manager; vault Owner is a multisig controlled by Business	Gates live funds
D4	Redemption service level	Publish an operational target (for example, within 5 business days, subject to liquidity)	Members see stages, not a guarantee
D5	Integrated top-up at launch	No; direct crypto funding	Removes a provider from the critical path
D6	Identity checks	Counsel decides the tier; build the gates now; screen addresses from day one	Avoids rework if KYC is required
D7	Points rules	No cash value, not transferable, not awarded for deposit amounts	Keeps Points separate from investment activity
D8	Members without a valid Sponsor	Attach to a company root account, flagged for review	Keeps the genealogy complete
D9	Alpha community import	Import before launch with original Sponsors and dates; reconcile counts and edges (GEN-03)	Existing capital and relationships arrive with history intact
D10	Placement window promised in the field call	Not in the Pilot; publish the policy first, then enable the governed workflow (GEN-04 to GEN-06)	Avoids building moves before the rules exist
D11	Payout treasury and approvers	Safe multisig owned by Business; Finance approves each batch; payouts only to the Builder's Helm wallet	Payout controls (PAY-01 to PAY-04)
D12	PillarsHub status	Confirmed per Business; validate it with Terrel's proof scenarios T01–T06 in staging, not a new selection	Reconciles Terrel's "candidate" wording with the confirmed stack
D13	MLM admin and Builder back office	Use the PillarsHub Portal and back office (SSO) instead of building them in Helm; Helm shows essentials only	Saves weeks of UI work; needs PillarsHub's Portal approvals and SSO token handling confirmed

PART II: TECHNICAL IMPLEMENTATION

9 Traceability to Terrel's requirements

Terrel's *HELM Community Technology Requirements v0.2* (6 October 2026) defines "Must" as required **before the named capability is enabled**, and states that a deferred capability stays disabled. That lets Phase 0 enable a small set of capabilities fully, rather than all capabilities partly. The table maps each requirement group to the Phase 0 Pilot.

Group	Requirement IDs	Phase 0 Pilot treatment	Where in this document
Identity and member records	ID-01 to ID-04	In scope. Stable person ID; Customer and Builder status; prospect, preregistered, verified and enrolled states; country eligibility and agreement versions	FR-1, FR-15; section 13
Corporate administration	ADM-01 to ADM-03	In scope. Separate view, configure, correct, approve and release roles; maker-checker on Sponsor corrections, holds and payout releases; test environments	FR-13, FR-19; section 11.3
Lifecycle actions	ADM-04	Merges and recovery in scope as admin processes; suspension, termination and re-entry disabled until enabled	Section 13.2
Sponsor and genealogy	GEN-01 to GEN-03	In scope. Referrer, sponsor (enroller) and placement recorded separately with effective dates; loop and duplicate checks; alpha import with reconciliation	FR-3, FR-4, FR-20; section 13
Placement window	GEN-04 to GEN-06	Disabled until the placement policy is approved (decision D10); the data model already preserves original referral and history	Section 13.2
Compensation	COMP-01 to COMP-04	In scope, simple plan only. One approved eligible event, versioned three-level Unilevel, replay-safe events, calculated, pending, held, payable and paid states, reconciliation to batches	FR-17, FR-18; sections 12, 13.4
Qualification and rank	QUAL-01 to QUAL-03	Disabled. No ranks in the simple plan; QUAL-03 applies only when rank progress is promised	Scope table
Earnings audit	AUD-01 to AUD-03	In scope. Explain each earning from source event to payout; preserve corrections; period statements separating vault performance from referral earnings	FR-18, FR-19
Rewards	REW-01 to REW-03	Points only; no redemption, promotional credit or cash conversion	FR-11; decision D7
Builder back office	BO-01, BO-02, BO-04	In scope. Essentials in Helm (referral link, team counts, earnings summary, payouts, dispute request); detailed statements and tree in PillarsHub's back office via SSO; mobile web and release language (PillarsHub back-office mobile behaviour TBD)	FR-16, FR-19
Team visibility	TEAM-01, TEAM-02	In scope, minimum. Direct team and counts; no access to another branch or anyone's balances, trades or support cases	FR-16
Team tools	TEAM-03 to TEAM-05	Basic welcome step only (TEAM-05 Pilot); search, contact and automation deferred	Scope table
Campaigns	CAMP-01	In scope. Referral link attribution, precedence and invalid referrals; CAMP-02 to CAMP-04 disabled	FR-3
Business intelligence	BI-01, BI-02	In scope. Separate definitions for customers, Builders, deposits, Commissions, liabilities and payouts; reconciliation to sources	FR-14
Payouts	PAY-01 to PAY-04	In scope. Payout instructions from approved payable amounts; restricted destinations; tracked states; reconciliation; payee status	FR-18; sections 12, 15

Group	Requirement IDs	Phase 0 Pilot treatment	Where in this document
APIs and events	API-01 to API-03	In scope. Sandbox, stable event IDs, replay and backfill, polling where webhooks are unsafe; export rights in contract	Sections 11, 12; launch gate
Support	SUP-01 to SUP-07	In scope (Pilot). Agent context view, case links to person and event IDs, agent limits, FAQs and runbooks, metrics	FR-12
Compliance records	REC-01, REC-02	In scope. Versioned agreements, consent and disclosures; country and product restrictions with rule version	FR-15; section 15.3
Operations	OPS-01 to OPS-04	In scope. Load envelope, recovery targets, incident response, per-capability gates	Section 15
Implementation	IMP-01 to IMP-04	In scope. Plan with owners, migration rehearsal, release acceptance, handover	Sections 14, 15

Where this document differs from Terrel's draft:

- **Provider status.** Terrel's draft lists Pillars and Infinite MLM Software as candidates, with neither selected. Business has since confirmed PillarsHub. This document treats PillarsHub as confirmed and proposes running Terrel's proof scenarios as **validation** of the configuration (decision D12), not as a new selection.
- **Proof scenarios for the Pilot.** T01 (alpha migration), T02 (plan normal and boundary cases), T03 (duplicate and replayed events), T05 (reversal before and after payout), T06 (payout timeout and retry), T07 (leader access limits) and T11 (Customer, Builder and dual role) gate the Pilot. T04, T09, T12 and T13 gate the capabilities they relate to when those are enabled. T10 and T14 gate production handover.
- **Still missing from Terrel's "Waiting for" list:** the compensation specification with worked examples, alpha enrollment and genealogy records, and a one-page release scope (audience, countries, languages, activities).

10 Refined high-level PRD

10.1 Objectives

1. Onboard Members, including the alpha community, with one stable identity and a usable, self-custodial wallet.
2. Let Members fund their wallet with crypto and join the approved vault, with a practical Redemption journey.
3. Capture the complete genealogy, with referrer, sponsor and placement kept separately, from the first signup.
4. Pay a simple three-level Commission on one approved eligible event, in Finance-approved USDC batches.
5. Reuse the existing Tap Prediction game and Points to drive engagement.

10.2 Personas

- **Member (Customer):** joins through a referral, funds a wallet, deposits, plays, redeems, asks for support.
- **Builder:** a Member enrolled to refer others; sees the referral link, direct team and Commission states; receives payouts. Not every Member is a Builder (ID-01).
- **Operations administrator:** reviews referral, import and funding exceptions, maintains the vault allowlist, monitors queues and reconciliation, handles escalations.
- **Finance approver:** reviews and approves payout batches and holds; signs treasury transactions.
- **Vault operator** (Business-appointed): updates the vault valuation, executes deposit and redemption queues, manages liquidity. Uses the Onyx Admin App.
- **Support agent:** answers Members in Chatwoot with read-only context; cannot move funds, change Sponsors, release payouts or award Points.
- **Business reviewer:** reads funnel, participation, referral, Commission and engagement reports.

10.3 Functional requirements and acceptance criteria

ID	Requirement	Acceptance criteria
FR-1	Authenticate with Privy and resolve one Helm member	Repeat logins with any linked method resolve to the same Helm member ID; the backend verifies every Privy token; duplicate-person cases go to an admin queue (ID-02)
FR-2	Create one embedded EVM wallet per Member	Wallet exists after first login; address is read from Privy server-side; the Member can sign a test transaction; recovery and export behaviour is explained in the app
FR-3	Capture a Referral Request at signup	Invite link captured before login and bound to the new member; attribution precedence and window applied (CAMP-01); self-referral rejected; missing or invalid Sponsor follows rule D8
FR-4	Accept the Sponsor Edge in PillarsHub	Customer created with the Helm-defined ID, signup date and enroller; node read back with the expected upline; edge marked accepted only after read-back; retries never create duplicates; original referrer kept in Helm (GEN-01)
FR-5	Show supported funding network and asset	Only one network and asset are offered; wrong-network warning is shown; funding moves through submitted → confirmed (N blocks) → credited; failed or unsupported transfers go to exceptions
FR-6	Screen addresses	Member wallet on creation, each funding source and each Redemption destination are screened; a hit blocks the vault allowlisting, payout or action and opens a case
FR-7	Gate vault access	Member must accept current vault terms (versioned) and pass eligibility checks; only then is the wallet added to the Onyx allowlist
FR-8	Vault deposit	Member signs exact-amount approval and deposit request; Helm tracks approval → request → pending → executed or cancelled; position counts only after execution
FR-9	Position display	Shares and value as of the vault's last valuation, with the valuation timestamp; matches chain state

ID	Requirement	Acceptance criteria
FR-10	Redemption	Member signs a redemption request; Helm shows pending, awaiting liquidity, executed, cancelled; asset arrives in the Member's wallet; position, business events and any Commission adjustment follow the approved rule
FR-11	Games and Points	Linked game account; verified events award Points once; rules and daily limits visible; Points shown separately from money (REW-01)
FR-12	Support	Verified chat identity; agent context view shows identity, Customer or Builder status, community status, earnings and payout state, and case history (SUP-01); cases link to person and event IDs (SUP-04); agent actions limited (SUP-05); FAQs and runbooks versioned (SUP-06); logout resets the chat
FR-13	Admin and reconciliation	Separate view, configure, correct, approve and release roles (ADM-01); daily reconciliation of genealogy, funding, vault positions, Commissions, payouts and Points; mismatches in an exception queue with audit trail
FR-14	Reporting	Separate definitions for customers, Builders, deposits, Commissions, liabilities and payouts, with cutoffs and freshness shown and no double counting (BI-01, BI-02)
FR-15	Customer and Builder status, preregistration	Prospect, preregistered, verified and enrolled states tracked separately; Builder enrollment explicit and versioned; preregistered people cannot receive earnings (ID-04); restricted capabilities blocked by country and agreement version (ID-03, REC-02)
FR-16	Team view	Builder sees referral link and direct-team counts with freshness in Helm, and opens PillarsHub's back office through SSO for the tree and reports; neither view exposes another branch or anyone's balances, positions, trades or support cases (TEAM-01, TEAM-02, BO-01; PillarsHub visibility settings to confirm)
FR-17	Post eligible events for Commission	Only the approved eligible event type is posted, after on-chain confirmation, with a stable external ID; replays create one obligation; reversals create a traceable adjustment (COMP-01, COMP-02)
FR-18	Simple payout	Released bonuses arrive as a PillarsHub batch at Helm's merchant endpoint; Finance approves; Helm proposes treasury transactions to the Builder's own Helm wallet only; each payment is reported back as Success, Failure or Pending with the transaction reference; a timeout and retry produce one payment or an explicit exception (PAY-01 to PAY-03, T06)
FR-19	Earnings statement and dispute	Builder sees an earnings summary and payouts in Helm, with detailed bonus statements in PillarsHub's back office; states shown are calculated, pending, held, payable and paid with reasons (COMP-03); period statement separates vault performance from referral earnings (AUD-03); a dispute request carries the earning ID (BO-02)
FR-20	Alpha community import	Alpha members imported Sponsors-first with original dates and provenance "import"; counts and edges reconcile; exceptions resolved or approved; rollback rehearsed (GEN-03, IMP-02, T01)

10.4 Operational requirements

- Every integration write is idempotent from Helm's side, retried with backoff and reconciled daily (API-02).
- No private key, seed phrase or privileged provider credential reaches the browser, logs or support payloads (SUP-03).
- Separate test and production environments for every provider, with separate credentials (ADM-03).
- Monitoring and alerts for: stale vault valuation, deposit or redemption requests pending beyond target, reconciliation breaks, sync failures, screening hits, payout batches not confirmed within the agreed window.
- An agreed load envelope (people, tree depth, events per hour) and recovery targets before production testing (OPS-01, OPS-03).

10.5 Success metrics (targets TBD)

Time to first approved live deposit and to first payout; signup completion; signup-to-funded and funded-to-deposited conversion; referral attribution completeness (target 100% of accepted signups); alpha

import reconciliation (target zero unexplained differences); payout batch accuracy; reconciliation exceptions per week; game participation; support first-response and resolution time.

11 Solution architecture

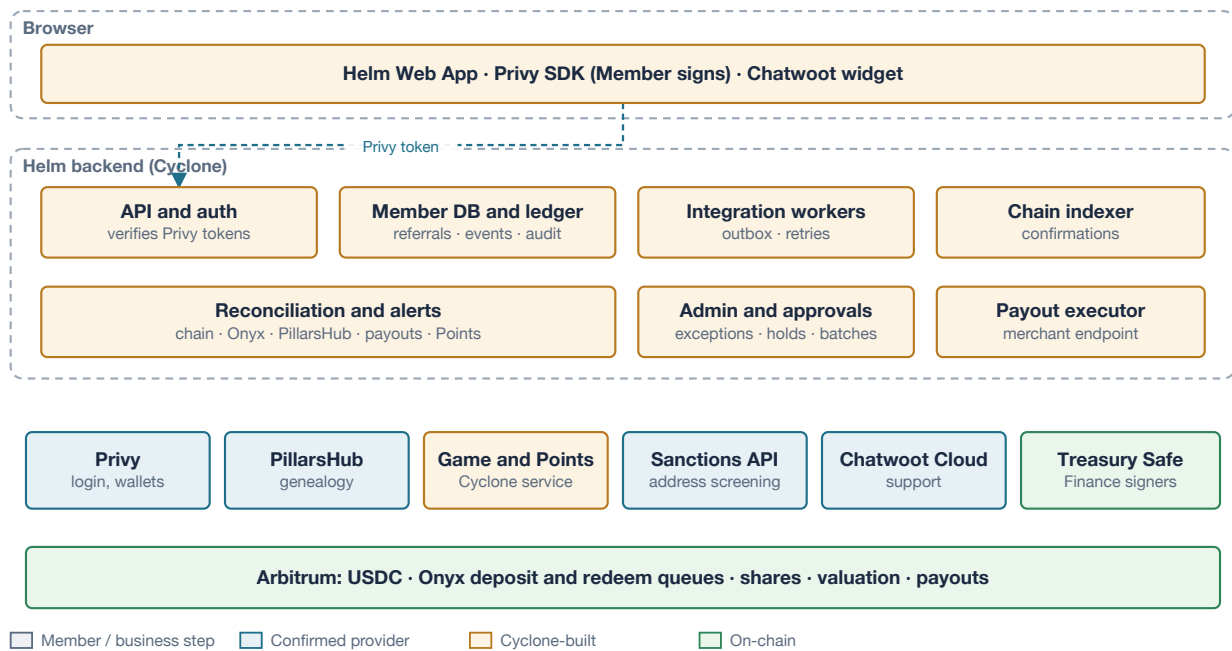


Figure 3: Technical architecture and trust boundaries.

11.1 Components and why each is needed

Component	Responsibility	Why custom
Helm Web App	Member and admin UI; Privy login; signing via the Member's embedded wallet; Chatwoot widget	One Member experience across providers
API and auth	Verifies Privy access tokens; resolves Helm member ID; serves the app	Providers cannot share one identity on their own
Member DB and ledger	Members, provider-ID mappings, wallets, Referral Requests, Sponsor Edges and corrections, business events, consent records, audit log	Single place to reconcile identities and money events
Integration workers	Outbox jobs to PillarsHub, game service, screening and allowlist; retries and dead letters	No provider offers idempotency keys across all calls
Chain indexer	Indexes USDC transfers to Member wallets and Onyx queue, shares and valuation events at confirmation depth	Privy production webhooks need Enterprise; Onyx has no webhooks. Verified
Reconciliation	Daily checks against chain, Onyx API, PillarsHub tree and Points	Provider data can drift; must be provable
Admin console and approvals	Helm-owned data only: exceptions, screening cases, allowlist status, import exceptions, payout execution and holds, with maker-checker; deep links to the PillarsHub Portal for genealogy, plan, bonuses and release	PillarsHub Portal covers MLM administration; Helm covers identity, wallets and money movement
PillarsHub Portal and back office (reused)	Genealogy browsing, Sponsor and placement edits, plan, bonuses, period close, release, MLM reports; Builder back office through SSO	Not custom: reused to save build time
Payout executor	Receives PillarsHub payout batches at a custom merchant endpoint; validates the callback token; checks payees and holds; proposes USDC transfers to the treasury multisig; reports Success, Failure or Pending per payment	PillarsHub calculates and releases but does not move crypto; no payment provider in scope
Alpha import tool	Loads alpha members Sponsors-first, reconciles counts and edges, reports exceptions, supports rollback	No bulk import API in PillarsHub. Verified

11.2 Data authority

Object	Authoritative record	Helm keeps
Helm member identity and provider-ID mapping	Helm DB	Master
Login identity	Privy user (DID)	Mapping only
Wallet address	Privy (server-side read) and chain	Association with first-seen time
Accepted genealogy	PillarsHub tree (customer enroller plus node upline)	Pending requests, accepted-edge copy, audit log of corrections
Wallet funding	Blockchain	Indexed, confirmed events
Vault deposit requests, positions, Redemptions	Blockchain (Onyx contracts)	Indexed events; Onyx API as cross-check
Vault valuation	Onyx valuation contract (set by vault operator)	Displayed with timestamp
Commission obligations (calculated, released)	PillarsHub bonuses and batches	Statement copy with source event IDs
Payout execution	Blockchain (treasury transfers)	Payout record with batch, payment and transaction IDs; status reported to PillarsHub
Customer and Builder status, preregistration	Helm DB	Master; mirrored to PillarsHub customer type and status
Points balance	Cyclone points service	Display copy
Conversations	Chatwoot	Link to member ID
Terms acceptance, screening results, eligibility	Helm DB	Master

11.3 Credentials, signing and custody boundaries

Actor	Can sign or act	Cannot	Notes
Member	Wallet transfers, vault approval, deposit and redemption requests, cancellations	Admin actions	Privy embedded wallet; Onyx requires the Member's own address to sign its requests. Verified
Helm backend	PillarsHub writes (customers, nodes, sources) with a limited token; game API; screening API; vault allowlist updates through a list-owner key; propose treasury payout transactions	Move Member funds; vault admin actions; execute a payout without Finance signatures	List-owner key held in a key management service; owns only the allowlist contract (to confirm with Enzyme)
Vault Owner (Business)	Add and remove Admins; any admin action	—	Recommended: Safe multisig with Business signers
Vault Admin / operator (Business)	Update valuation; execute queues; move assets to the strategy wallet; set fees	—	Fully trusted by the protocol. Verified
Treasury signers (Finance)	Approve and sign payout transactions from the treasury multisig	Change payout destinations	Payouts go only to the Builder's Helm wallet; a threshold of Finance signers is required
Strategy Manager (Business-appointed)	Runs the strategy outside Onyx; returns liquidity	—	Custody of the strategy wallet sits outside Onyx. Verified
Enzyme	Deploys; upgrades contracts (global owner)	—	Upgrade governance must be in the MLA. Verified
Privy	Runs the signing enclave	Sign for the Member	Privy "is never an authorized signer". Verified
Support agent	Reply in Chatwoot; read-only member status	Funds, Sponsors, Points	Enforced in Helm admin

PillarsHub tokens are environment-bound and do not expire; they can be made read-only. Recommended: a read-only token for reconciliation and a separate write token, with destructive areas denied if PillarsHub allows it (TBD). Bonus release and batch creation are Finance actions in the PillarsHub Portal, not calls from Helm's backend; Helm only receives the resulting batch and reports payment status.

12 End-to-end data flows

Every flow distinguishes **submitted** (Helm has a request or transaction hash), **confirmed** (provider acknowledged or block confirmations reached) and **synchronised** (Helm's copy matches the authority after reconciliation). Webhooks are treated as hints; polling and indexing are authoritative.

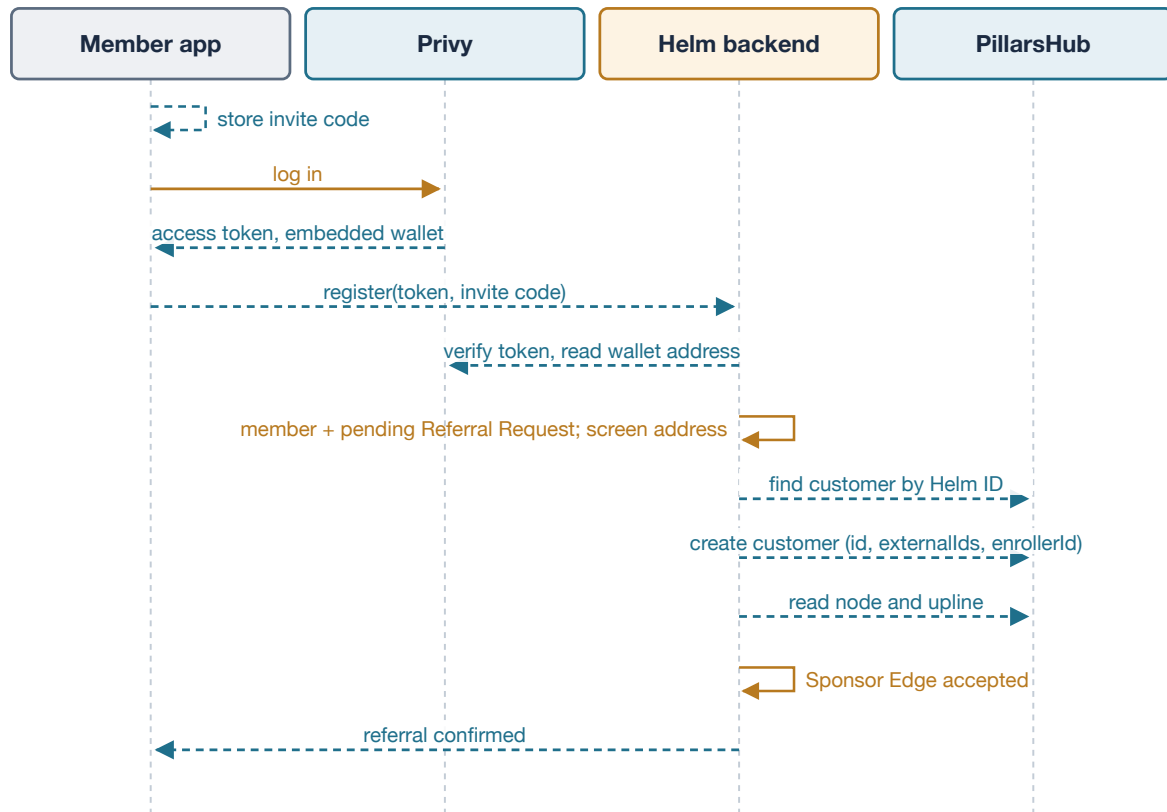


Figure 4: Invite, signup, wallet and accepted genealogy.

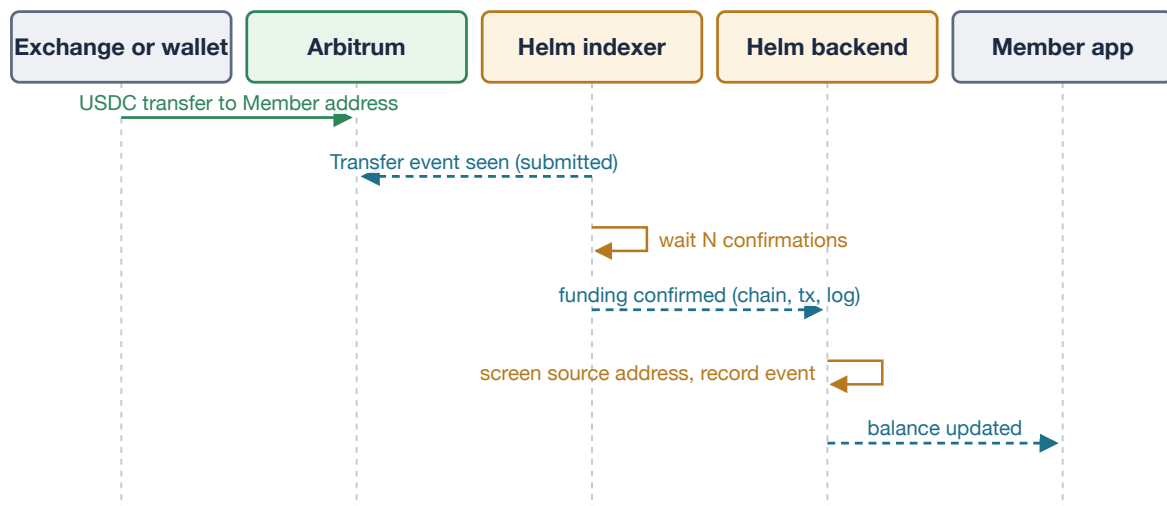


Figure 5: Direct wallet funding.

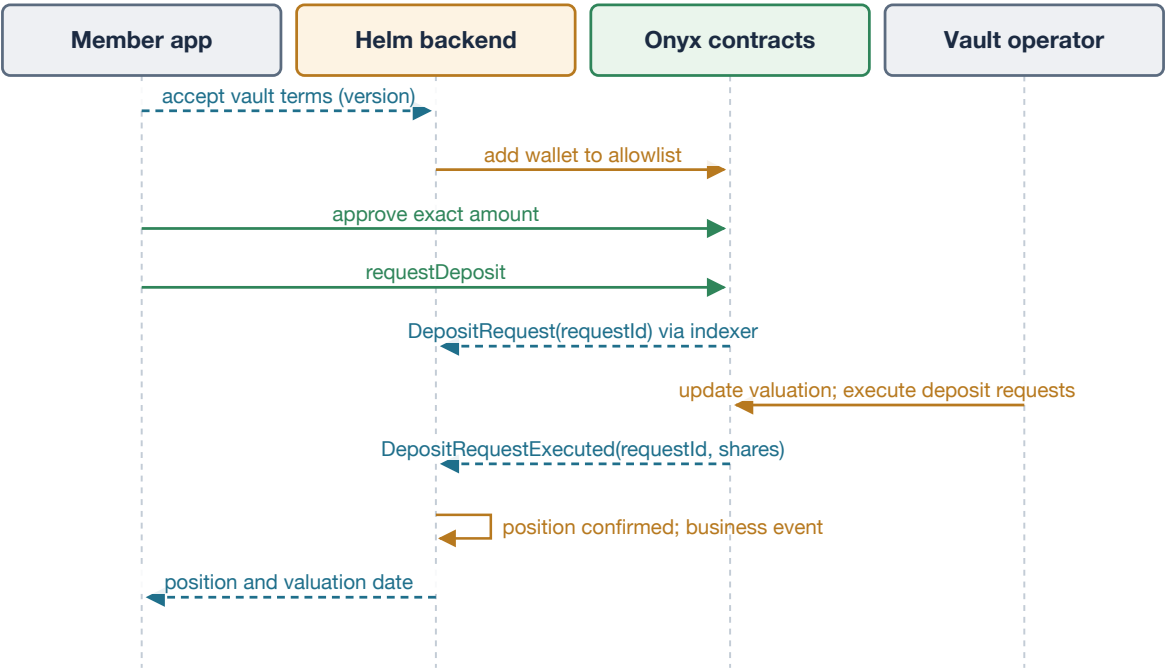


Figure 6: Vault deposit, execution and position.

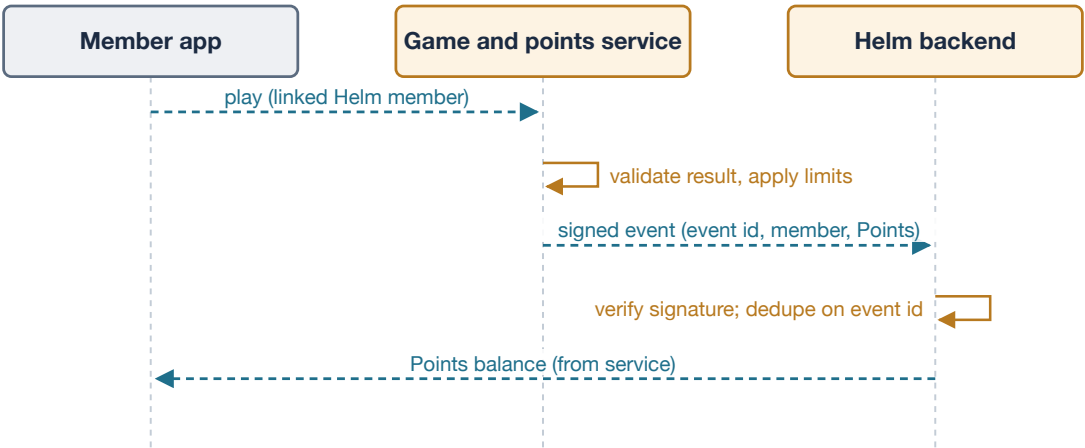


Figure 7: Game activity to Points.

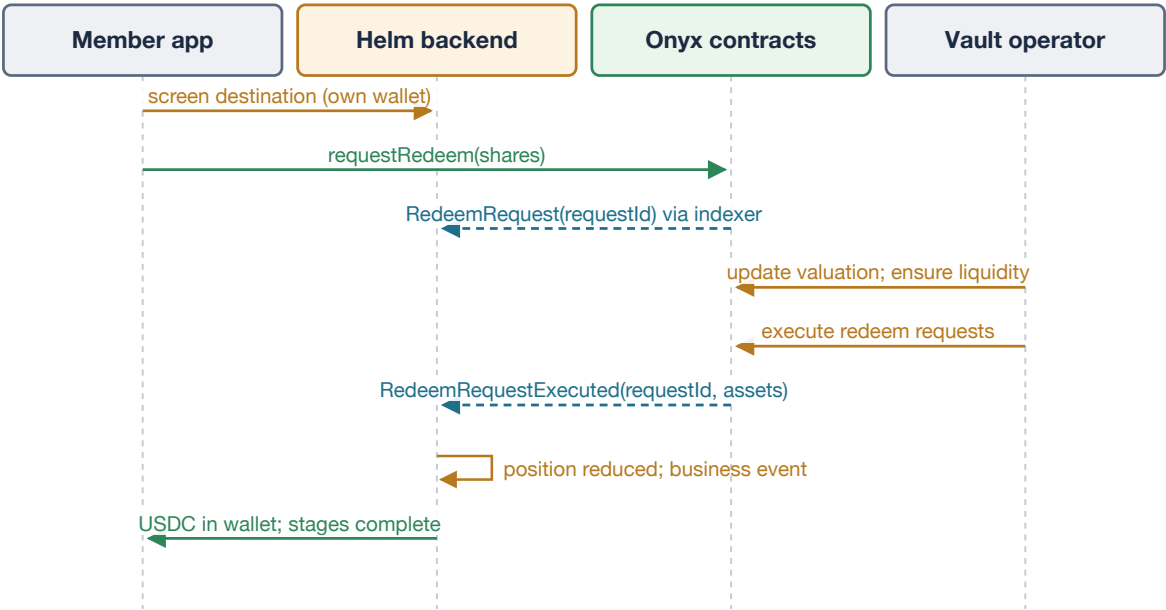


Figure 8: Redemption request to settlement.

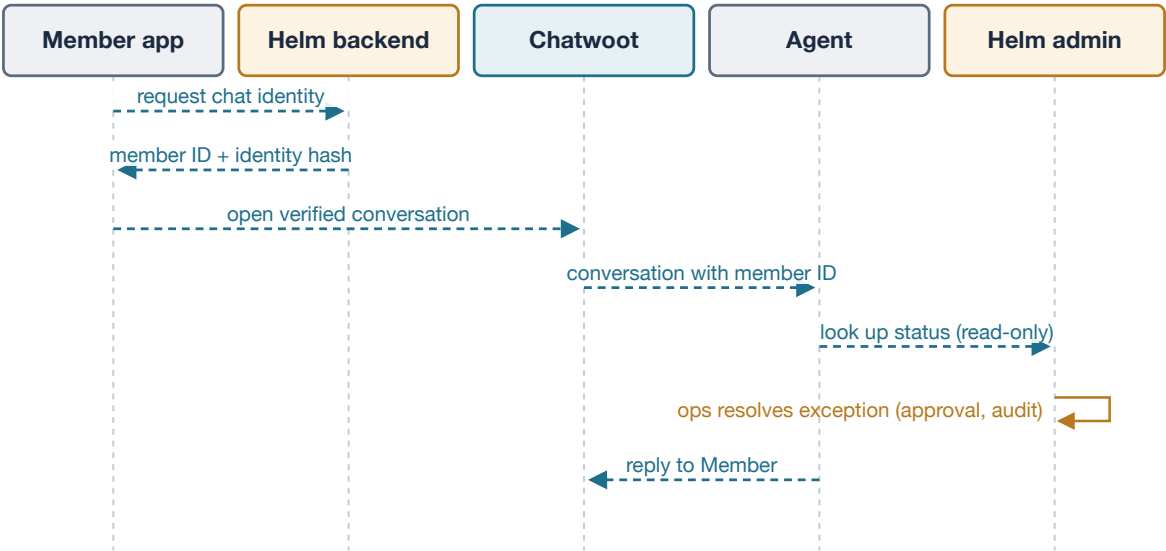


Figure 9: Support and admin exception resolution.

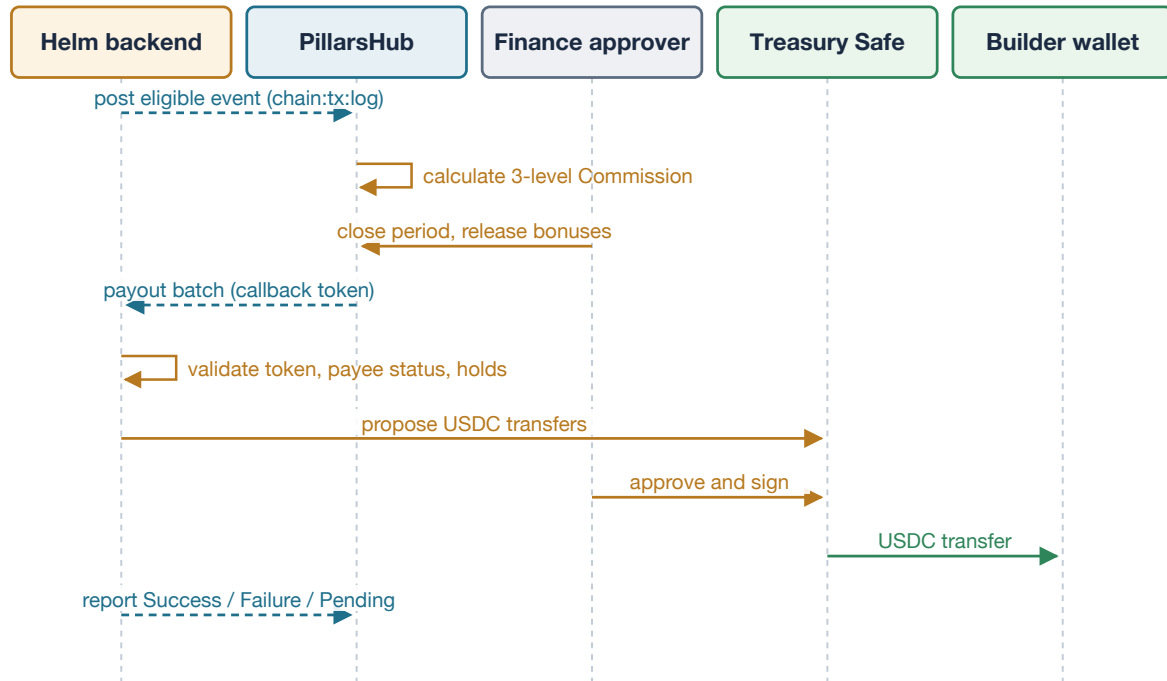


Figure 10: Eligible event to Commission and simple payout.

12.1 Flow matrix

Flow	Origin → destination	Minimum data	Authoritative record	Deduplication key	Failure and retry	User sees
Signup and referral	App → Helm → PillarsHub	Privy DID, wallet address, invite code, Sponsor member ID	PillarsHub tree (accepted edge)	Helm member ID (PillarsHub customer ID)	Look up before create; backoff; dead letter to ops; nightly tree reconciliation	“Referral pending” then “confirmed”
Wallet funding	Exchange → chain → Helm indexer	Chain, tx hash, log index, amount, from, to	Chain	(chain, tx hash, log index)	Re-index from last safe block; reorg handling by confirmation depth	Pending → confirmed balance
Vault deposit	App → Onyx contracts; indexer → Helm	Request ID, amount, wallet, terms version	Chain (queue and shares)	(chain, queue address, request ID)	Reverted transactions shown with reason; stale pending alert; Onyx API cross-check	Approval → requested → processing → position
Eligible event (approved basis only)	Helm → PillarsHub sources	Member node, amount, business time, receipt time, settlement state, rule version	Helm business event; PillarsHub source	Source externalId = chain:tx:log	Look up by externalId on conflict or timeout; reversal posts a linked adjustment	Builder sees Commission as pending
Commissioner payout	PillarsHub → Helm merchant endpoint → treasury → chain	Batch ID, payment ID, node, bonus, amount, currency, period	PillarsHub (obligation); chain (payment)	Batch ID plus payment ID; transaction hash	Batch ID is the same on retries; Helm never pays a payment ID twice; unconfirmed transfers stay Pending; failures reported per payment	Pending → held → payable → paid, with transaction link
Game to Points	Game service → Helm	Event ID, member ID, Points, rule ID	Points service	Event ID	Signed events; replay protection; daily limits	Points balance and history

Flow	Origin → destination	Minimum data	Authoritative record	Deduplication key	Failure and retry	User sees
Redemption	App → Onyx; indexer → Helm	Request ID, shares, destination	Chain	(chain, queue address, request ID)	Awaiting-liquidity alert; cancellation after minimum duration	Requested → awaiting liquidity → paid
Support	App → Chatwoot; agent → Helm admin	Member ID, identity hash, non-sensitive status	Chatwoot (conversation); Helm (actions)	Conversation ID	Widget reset on logout; signed Chatwoot webhooks logged	Chat thread

Unavailable providers: if PillarsHub is down, signups proceed with the referral pending and sync later. If the indexer lags, balances show “updating”. If Onyx’s API is unavailable, Helm relies on chain reads. If the game service is down, play is disabled and no Points are lost.

13 Genealogy, Commission and Phase 1 readiness

13.1 Minimum data model

Table	Key fields	Purpose
member	helm_member_id, handle (optional, proposed by Terrel), lifecycle_state (prospect, preregistered, verified, enrolled), is_builder, builder_enrolled_at, country, created_at	Stable identity; Customer and Builder distinction (ID-01, ID-04)
provider_identity	helm_member_id, provider (privy, pillarshub, chatwoot, game), external_id, linked_at	Explicit mappings
wallet	helm_member_id, chain_id, address, source (embedded), first_seen_at, screening_status, allowlist_status	One vault and payout wallet per Member
agreement_acceptance	helm_member_id, agreement (member terms, Builder terms, vault terms, privacy), version, accepted_at	Versioned consent (REC-01)
referral_request	id, helm_member_id, referrer_member_id, invite_code, channel, captured_at, status, rejection_reason	Pending and rejected referrals; original referrer (GEN-04)
sponsor_edge	id, member_id, sponsor_member_id (enroller), placement_upline_id, effective_at, accepted_at, provenance (signup, import, correction), policy_version, pillarshub_sync_status, superseded_by	Full graph at every depth; sponsor and placement kept separately (GEN-01)
edge_correction	id, edge_id, old values, new values, reason, requested_by, approved_by, approved_at, consent_ref	Audited changes (ADM-02)
business_event	id, type, member_id, amount, currency, business_time, receipt_time, settlement_state, product, country, attribution, reverses_event_id, rule_version, source_ref, idempotency_key	Terrel's minimum event contract
commission_statement	helm_member_id, period, bonus lines with PillarsHub IDs, state (calculated, pending, held, payable, paid), payout_ref	Builder statements (COMP-03, AUD-03)
payout	batch_id, payment_id, helm_member_id, amount, currency, status (received, approved, proposed, signed, confirmed, failed), safe_tx_hash, chain_tx_hash, reported_at	Payout tracking (PAY-02)
import_run	run_id, source, counts, exceptions, approved_by, rolled_back	Alpha import evidence (IMP-02)
integration_outbox	id, target, payload hash, attempts, status, last_error	Reliable sync

Only data with an identified purpose is collected. No Sponsor or member data is written on-chain.

13.2 Rules

- **Customer versus Builder:** every Member starts as a Customer; Builder enrollment is an explicit, versioned step with Builder terms. Preregistered people keep their attribution but cannot earn (ID-04).
- **Self-referral:** rejected at capture (same member, same wallet, same verified email or phone).
- **Invalid or missing Sponsor:** attach to the company root account with provenance “no sponsor” and flag for review (decision D8).
- **Cycles and duplicates:** impossible for new Members; checked on every correction and import (GEN-02).
- **Sponsor and placement changes:** never self-service in the Pilot. Made by authorised admins in the PillarsHub Portal, with reason and consent reference recorded in a Helm case. Helm detects every change through Node/Updated webhooks and nightly reconciliation, and opens a case for any change without an approved request. Whether the Portal itself enforces a second approver and keeps an audit log is not verified; if it does not, the approval is recorded in Helm before the Portal edit (ADM-02). The governed placement window (preview, consent, deadline, notifications; GEN-04 to GEN-06) is enabled only after the policy is approved (decision D10). Customer movements stay disabled in the PillarsHub tree so the two systems cannot drift.

- **Account merges:** admin process; keep the earliest accepted edge; re-point child edges with audit entries; retire the duplicate identity (ADM-04).
- **Alpha community import:** Sponsors before children, with original signup dates and provenance “import”; record by record because there is no bulk API. Verified. Reconcile counts and edges, resolve or approve every exception, and rehearse rollback before cutover (GEN-03, IMP-02).

13.3 Proving correctness

- **Full export:** page through every node of the tree with an as-of date, not the downline endpoint, which is capped at 10 levels. Verified.
- **Nightly reconciliation:** compare Helm’s accepted edges with PillarsHub nodes (count, edge-by-edge upline match, checksum). Any mismatch is an exception.
- **Staging tests before launch:** a synthetic tree at least 15 levels deep; the alpha import dry run (T01); Terrel’s scenarios T02, T03, T05, T06, T07 and T11 with Finance-prepared expected results.
- **History:** PillarsHub accepts an as-of date and keeps dated placements, and Helm keeps effective timestamps, so the tree on an original earning date can be shown after a later correction (GEN-01 proof).

13.4 Simple Commission and payout

- **Plan.** Use the three-level Unilevel plan already connected in PillarsHub staging (Correspondence). PillarsHub configures the approved rates; the plan is read-only through the API, so Helm cannot change economics. Verified.
- **Eligible event.** Helm posts only the approved event type (decision D2) to a dedicated source group, after on-chain confirmation, with external ID `chain:tx:log`. PillarsHub calculates in real time, so nothing is posted until the basis is approved. Verified.
- **Reversals.** If the approved rule adjusts Commissions on Redemption or failed events, Helm deletes the source or posts a linked adjustment before release; after payout, the agreed recovery rule applies (T05).
- **Holds and release.** Builders without required payee status or with a screening hit are held. Finance closes the period, reviews and releases bonuses in PillarsHub.
- **Execution.** PillarsHub sends the batch to Helm’s merchant endpoint with a callback token, which Helm validates against PillarsHub. Helm proposes USDC transfers to each Builder’s Helm wallet in the treasury Safe; Finance signers approve; Helm reports each payment’s status back. Verified (batch interface).
- **Reconciliation.** Daily: eligible events → PillarsHub sources → released bonuses → batch payments → on-chain transfers → statements (COMP-04).

14 Backlog, dependencies and schedule

14.1 Prioritised backlog

Priority	Work items	Depends on
Start immediately	App shell; Privy login and embedded wallet; member DB with Customer and Builder states; referral capture and validation; PillarsHub staging sync with read-back; alpha import tool on sample data; event pipeline and source posting (staging only); payout merchant endpoint and Safe proposals on testnet; chain indexer; Onyx deposit and redemption on Sepolia; Chatwoot widget; admin skeleton with roles; screening hook; CI, environments, secrets	Provider test access; alpha sample data
Thin slice (weeks 1–3)	End-to-end demo on test networks: signup → referral → deposit → test Commission → test payout batch → Redemption; one game-to-Points flow	Game API docs
Before live funds and payouts	Vault terms and disclosures; allowlist automation; Builder essentials and PillarsHub SSO hand-off; Builder enrollment and terms; payout approvals, holds and reporting; alpha import rehearsal and reconciliation; support agent view, FAQs and runbooks; reconciliation jobs and alerts; country eligibility; production environments; load and recovery tests; security review fixes; operations rehearsal	Strategy, MLA, ownership, Commission basis, legal answers
Deferrable (disabled)	Governed placement window; ranks and qualification; campaigns; integrated top-up; additional chains and assets; Telegram and WhatsApp; commercial transaction monitoring; native mobile	Business demand and approved rules

14.2 Critical path

1. Meeting decisions D1–D13.
2. **Vault readiness:** strategy and Manager → vault configuration (asset, async queue, external allowlist, non-transferable shares, fees) → MLA signature → production deployment → ownership handover to the Business multisig.
3. **Commission basis and plan proof:** approved eligible event, rates, period and hold → PillarsHub configuration → Finance expected results match (T02).
4. **Alpha data:** records received → dry run → reconciliation → approved exceptions → rehearsed cutover (T01).
5. **Legal answers:** entity, launch countries, KYC tier, vault terms, referral payout model.
6. **Security review** of Helm's integration, payout executor and vault configuration.
7. **Operations rehearsal:** valuation, queue execution, Redemption liquidity and a small payout batch.
8. PillarsHub production launch slot (Tuesday to Thursday).

Engineering runs in parallel with items 2–5; the launch date is set by whichever finishes last.

15 Launch and operational readiness

15.1 Gate for accepting real funds and paying Commissions

All must be true; none is waived to meet a date.

1. Vault strategy, Manager, fees and Member terms approved by Business and counsel.
2. MLA signed; deployed version and audit coverage confirmed by Enzyme; upgrade governance documented.
3. Vault Owner multisig in place; Admin and list-owner keys in managed custody; no Owner or Admin key in Helm's backend.
4. Deposits and Redemptions proven on production with a small operator-funded amount.
5. Daily reconciliation passing; monitoring and alerts live; incident contacts at each provider.
6. Screening and eligibility live for the counsel-approved tier.
7. Security review completed and high-severity findings fixed.
8. Written provider acceptance of the business model (Privy, Enzyme, PillarsHub).
9. Support agent view, FAQs, runbooks and escalation procedure, including handling of flagged users and earnings disputes (SUP-01 to SUP-06).
10. Alpha import reconciled, exceptions resolved or approved, rollback rehearsed (GEN-03, IMP-02).
11. **Before the first payout:** Commission basis, rates, period and hold approved by Owen, Finance and counsel; Finance-prepared plan examples match PillarsHub results (T02); duplicate, replay and reversal scenarios pass (T03, T05); a test batch with a forced timeout produces one payment (T06); treasury Safe with Finance signers in place; payee status captured (PAY-04).
12. Load envelope and recovery targets agreed and tested (OPS-01 to OPS-03); full data export rights in the PillarsHub and Enzyme contracts (API-03).

15.2 Security boundaries

- **Audit coverage.** ChainSecurity audited the Onyx core (December 2025), the cross-chain wallet (May 2026) and the Chainlink compliance integration (July 2026). One deployment helper added after the last audit appears in no audit scope found. Verified. Enzyme must confirm which components Helm's vault uses and that they are covered.
- **Upgrade risk.** Enzyme's global owner can upgrade all vault contracts; the audit describes this role as able to "fully drain the system", and no timelock was found. Verified. The MLA must cover notice and governance.
- **Gas sponsorship.** Privy's native sponsorship upgrades Member wallets with EIP-7702; the delegation contract is part of the security review. Verified.
- **Unsigned PillarsHub webhooks.** Treat them as hints and re-fetch through the API. Verified.
- **PillarsHub SSO.** Helm's backend mints a PillarsHub user token and opens the back office with the token in the URL query string. Verified. Tokens in URLs can leak through browser history and logs; confirm token lifetime and single use with PillarsHub, and open the back office in a new tab without referrer headers.
- **Payout callbacks.** PillarsHub authenticates payout batches with a callback token that Helm validates against PillarsHub's token endpoint before acting. Verified. The payout executor holds no signing key; it only proposes Safe transactions.
- **Test and production separation** for every provider, with separate credentials and no production keys on developer machines.

15.3 Compliance responsibility matrix

Accountable functions are proposals; named owners are TBD.

Control	Applicability question	Accountable (proposed)	Cyclone work	Provider capability relied on	Launch impact
Entity and custody classification	Does any entity have “control” of Member assets (vault roles, allowlist key)?	Business + counsel	Document signing authority and key flows	Privy self-custody (verified); Onyx roles (verified)	Blocker
Launch countries and geographic restriction	Which countries are allowed?	Business + counsel	Country attestation, IP checks, block list	None native	Blocker
Vault product characterisation	Is the vault offering a regulated product in launch countries?	Business + counsel	Terms, fee and risk screens with acceptance record	Onyx strategy and fees (TBD)	Blocker
KYC tier	Is any no-KYC tier allowed; at what thresholds?	Business compliance + counsel	Tier fields and gates now; vendor integration if required	None native (Privy KYC via Bridge unavailable)	Before funds
Sanctions: addresses	Which lists and when?	Business compliance	Screen wallet, funding source, Redemption destination	Chainalysis free API (verified, discovery)	Before funds
Sanctions: persons	Is name screening required?	Business compliance + counsel	Hook into identity tier	Vendor TBD	Before funds if required
Vault admission	Only eligible Members may deposit	Business + Enzyme	Allowlist automation after checks	Onyx deposit allowlist (verified)	Before funds
AML monitoring and escalation	What monitoring and reporting duties apply?	Business (compliance officer)	Thresholds, exception queue, case log	None native	Before funds (manual minimum)
Flagged users and transactions	What may Helm lawfully refuse or hold?	Counsel + Business	Restricted states in Helm; no assumed freezing power	Onyx admin discretion and policies (verified)	Before funds
Terms, privacy, consent, retention	Which notices and data-sharing consents?	Business (legal)	Versioned consent; data map per provider	Chatwoot Cloud hosted in the US (verified)	Before funds
Referral model and Points	Are referral rewards and Points disclosed and lawful?	Business + counsel	Rules pages; no earnings claims; Points separate from money	PillarsHub, game service	Before funds (disclosures)
Referral payout basis	Is paying Commissions on the chosen eligible event lawful in launch countries?	Owen + Finance + counsel	Configurable event pipe; nothing posted until approved	PillarsHub plan (verified)	Blocker for payouts
Payee status and tax reporting	What payee information is required before payout?	Finance + counsel	Payee status gate; hold process (PAY-04)	None native	Before payouts
Earnings disclosures	What earnings and income disclosures are required?	Business + counsel	Statements and disclosure screens (AUD-03)	—	Before payouts
Provider acceptance	Do providers accept the model on full disclosure?	Business	Avoid Stripe and Bridge features in Privy	Written confirmations	Before funds

16 Implementation decisions and feasibility gaps

Decision	Recommended answer	Evidence	Unresolved dependency	Responsible function (proposed)	Launch impact
Network and asset	Arbitrum One, native USDC; Sepolia for tests	Onyx and Privy support Arbitrum; Onyx test deployment is on Sepolia (verified)	Enzyme confirms production chain and asset	Business + Tech	High
Wallet model	User-owned Privy embedded wallet, no app signer, sponsored gas, export enabled	Privy docs (verified)	Privy accepts the business model; volume limit above \$1M per month	Tech	Medium
Vault configuration	Async deposit queue; external allowlist owned by a dedicated list-owner key; non-transferable shares; one asset	Onyx contracts and docs (verified)	Enzyme confirms external list ownership at deployment	Business + Enzyme	High
Vault roles	Business Safe multisig as Owner; separate Admin wallet; limited admin for valuation updates	Onyx roles (verified)	Business signers; Manager appointment	Business	Blocker
Redemption target	Publish an operational target; show stages	No protocol SLA (verified)	Liquidity policy from Manager	Business	High
Genealogy authority	PillarsHub, with Helm pending requests and reconciled copy	Customer and tree APIs (verified)	Auto-placement and correction method in staging	Tech + PillarsHub	Medium
Simple Commission	Three-level Unilevel on one approved eligible event; Finance-approved USDC batches from a Safe treasury	PillarsHub plan, sources and custom merchant batches (verified)	Approved basis, rates, period and hold; counsel	Owen + Finance + counsel	Blocker for payouts
Alpha import	Import before launch with history; reconcile and rehearse	Customer create with ID, date and enroller; no bulk API (verified)	Alpha records	Business + Tech	High
Placement window	Disabled in the Pilot; enable after policy approval	GEN-04 to GEN-06	Placement policy	Terrel + Tim	Medium
Customer and Builder	Separate states; explicit Builder enrollment	ID-01, ID-04	Builder terms	Terrel + Owen	Medium
Funding path	Direct crypto; no top-up provider	Paymenture publishes no assets, countries or fees and bars “pyramid schemes” (verified)	Business requirement	Business	Low
Points	No cash value; not linked to deposit amounts	No game docs supplied	Game API documentation; counsel review	Business + Cyclone	Medium
Identity checks	Gates built now; tier set by counsel; address screening from day one	No KYC in stack (verified)	Counsel decision	Business + counsel	High

What would change these defaults: integrated top-up at launch adds a payment provider to the critical path; ranks, matching bonuses or a placement window at launch add plan configuration, proof scenarios and operating capacity; counsel requiring full KYC adds an identity vendor and a gate before vault access and payouts.

Appendix A: Evidence

All links were read on 8 October 2026.

PillarsHub: [API authentication](#) · [Access tokens](#) · [Swagger UI](#) · [Customer API spec](#) · [Commission API spec](#) · [Webhooks](#) · [Webhook topics](#) · [Invoice dates and real-time commissions](#) · [Launch timing protocol](#) · [Money-out integrations](#) · [Terms](#)

Privy: [Pricing](#) · [Security FAQ](#) · [Supporting users](#) · [Webhooks](#) · [React quickstart](#) · [Acceptable use policy](#)

Enzyme Onyx: [Product and pricing](#) · [Architecture](#) · [Deposit control and allowlists](#) · [Redemptions](#) · [User roles](#) · [Deployments and upgrades](#) · [Risks and limitations](#) · [SDK](#) · [Public API](#) · [Contracts](#) · [Audits](#)

Chatwoot: [Developer docs](#) · [Pricing](#)

Funding and compliance: [Paymenture](#) · [Bridge developer agreement](#) · [Chainalysis sanctions oracle](#) · [FATF 2021 guidance](#)

Detailed notes with every source and quotation are in `work/notes/01-pillarshub.md`, `02-enzyme-onyx.md`, `03-privy.md` and `04-chatwoot-funding-compliance.md`.

Appendix B: Assumptions and gaps

- **Terrel's requirements v0.2 are traced in section 9.** Still missing from its "Waiting for" list: the compensation specification with worked examples, alpha enrollment and genealogy records, and a one-page release scope (audience, countries, languages, activities). Owners in Terrel's draft (Tim, Darren, JT, Finance, compliance) are proposals and are not repeated as commitments here.
- **Commission basis:** not yet approved. The design supports any single eligible event type; nothing is posted until Owen, Finance and counsel approve it.
- **Game and Points:** no API documentation was supplied. Needed: account linking method, event schema and signing, duplicate and replay protection, abuse controls, Points rules engine, whether Points can be redeemed for anything.
- **Not verified with providers:** PillarsHub auto-placement, depth limits, sponsor correction method, token permission granularity, rate limits, 409 semantics, pricing and security certifications; Privy paid-tier volume limits and business-model acceptance; Enzyme lead times, deployed version, external allowlist ownership, commercial terms; Chatwoot webhook retry behaviour.
- **Pricing:** Enzyme's public price differs from the correspondence; PillarsHub pricing is not public.
- **Team capacity, budget and launch cohort** were not supplied; schedule ranges assume about five to six engineers.
- **Placement-window expectations** from Nic's field call are not yet a published policy; the window stays disabled until it is.

Appendix C: Questions for providers

PillarsHub

1. Does creating a customer with an enroller place the node in the Unilevel tree automatically, with the upline equal to the enroller?
2. Is genealogy depth unlimited? Does listing tree nodes without node IDs return every node, and what is the page-size maximum?
3. What is the supported, audited way to correct a Sponsor? Can customer movements be disabled?
4. What does a 409 on posting a source mean? Is externalId unique per environment? Are duplicate customer IDs rejected?
5. Can a source group exist that no bonus uses? Is backdated volume accepted for closed periods?
6. Can access tokens be limited by area and deny deletes? What are the rate limits?
7. Can webhooks be signed? Is there a placement or source topic?
8. Please confirm in writing that API-based reconciliation of genealogy is permitted under the Terms, and explain the GLBA disclaimer in this context.
9. Security documentation, data residency, pricing and launch-slot booking.
10. Custom money-out merchant: can the batch carry a custom currency code such as USDC; retry behaviour when Helm reports Pending; how Failure is re-released; per-payment limits?
11. Can Helm's three-level Unilevel staging plan be configured with Finance's rates, period, hold and minimum, and who changes it (the plan is read-only through the API)?
12. Customer types or statuses for Customer versus Builder, and for preregistered people who must not earn.
13. Does the Portal support role-based approvals (second approver) and an audit log for Sponsor and placement changes, bonus edits and releases?
14. SSO user tokens: lifetime, single use, and whether a POST-based or short-lived exchange is available instead of a token in the URL. Can back-office visibility be limited so a Builder never sees another branch or balances, and how does the back office behave on mobile?

Privy

1. Do you accept this business model on full disclosure, and do embedded wallets, gas sponsorship and token verification require any Stripe or Bridge account?
2. How is the \$1M monthly transaction volume measured, and what applies above it on paid plans?
3. Can production webhooks or an SLA be added without Enterprise?
4. Security review material for the EIP-7702 delegation used by gas sponsorship.

Enzyme

1. Upgrade governance: who controls upgrades, with what notice, and can Helm's vault opt out?
2. Deployed version and addresses; audit coverage of every component Helm uses, including the deployment helper.
3. Confirm Sepolia for the test vault, Admin App access there, and API indexing of Sepolia.
4. Production chain and recommended deposit asset for Arbitrum.
5. Can an external allowlist owned by a Helm list-owner key be attached at deployment? Indexer delay?
6. Configure shares as non-transferable.
7. Redemption: notice-period or gate modules; recommended cadence; handling a batch that reverts on one request.
8. Valuation automation: lead time, cost, data-provider subscriptions.
9. Commercial terms (deployment fee, AUM option vs public plan, \$6,000 minimum), MLA draft, onboarding form, lead times.
10. Can Enzyme's default investor interface be disabled so Members use only Helm?

Chatwoot

1. Webhook retry and failure behaviour.
2. Data residency options for Cloud; audit logs below Enterprise.

Cyclone game team

1. API documentation, event signing, replay protection, abuse controls and Points rules engine.